ECON 4720: HA03

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Problems

Problem 1

Suppose that $Y_1, Y_2, \ldots, Y_n$ are random variables with a common mean μ_Y, a common variance σ_Y^2, and the same correlation ρ (so that the correlation between Y_i and Y_j is equal to ρ for all pairs $i \neq j$).

1a

Show that $\text{Cov}(Y_i, Y_j) = \rho\sigma_Y^2$ for $i \neq j$.

Solution Attempt: We know that correlation is defined as

$$Corr(Y_i, Y_j) = \frac{Cov(Y_i, Y_j)}{\sqrt{Var(Y_i)Var(Y_j)}}$$

and from our preamble we also know that "the correlation between Y_i and Y_j is equal to ρ for all pairs $i \neq j$" and common variance implying $Var(Y_i) + Var(Y_j) = \sigma_Y^2$. Thus,

$$\begin{aligned}
\rho &= \frac{Cov(Y_i, Y_j)}{\sqrt{\sigma_Y^2 \sigma_Y^2}} \\
&= \frac{Cov(Y_i, Y_j)}{\sigma_Y^2} \\
\rho\sigma_Y^2 &= Cov(Y_i, Y_j)
\end{aligned}$$

which shows what the question asks.

$\square$

1b

Suppose that $n = 2$. Show that $\text{E}(\bar{Y}) = \mu_Y$ and $\text{Var}(\bar{Y}) = \frac{1}{2}\sigma_Y^2 + \frac{1}{2}\rho\sigma_Y^2$.

Solution Attempt: This change brings our random variable set to $\{Y_1, Y_2\}$. First, we find $E(\bar{Y})$. Note that $\bar{Y}$ is

$$\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i = \frac{1}{2}(Y_1 + Y_2)$$

Thus,

$$E(\bar{Y}) = E\left(\frac{1}{2}(Y_1 + Y_2)\right) = \frac{1}{2}(E(Y_1) + E(Y_2))$$

From the preamble, we know that $E(Y_i) = E(Y_j) = \mu_Y$, thus,

$$E(\bar{Y}) = \frac{1}{2}(2\mu_Y) = \mu_Y$$

Now, for the variance, $Var(\bar{Y})$. From earlier, we know

$$\begin{aligned}
Var(\bar{Y}) = Var\left(\frac{1}{2}(Y_1 + Y_2)\right) &= \frac{1}{4}Var(Y_1 + Y_2) \\
&= \frac{1}{4}(Var(Y_1) + Var(Y_2) + 2Cov(Y_1, Y_2)) && \text{Expansion} \\
&= \frac{1}{4}(2\sigma_Y^2 + 2\rho\sigma_Y^2) && \text{From preamble and (a)} \\
&= \frac{1}{2}\mu_Y^2 + \frac{1}{2}\rho\sigma_Y^2 && \text{Shows what we want}
\end{aligned}$$

□

1c

For $n \geq 2$, show that E$(\bar{Y}) = \mu_Y$ and Var$(\bar{Y}) = \frac{\sigma_Y^2}{n} + \left(\frac{n-1}{n}\right)\rho\sigma_Y^2$.

Solution Attempt: Now our set of random variables is {Y_1, Y_2, ..., Y_n}. The expectation of $\bar{Y}$ follows a similar process as with (b).

$$\begin{aligned}
\bar{Y} &= \frac{1}{n}(Y_1 + Y_2 + \cdots + Y_n) = \frac{1}{n} \cdot n\mu_Y = \mu_Y \\
E(\bar{Y}) = E\left(\frac{1}{n}(Y_1 + Y_2 + \cdots + Y_n)\right) &= \frac{1}{n}(\underbrace{E(Y_1) + E(Y_2) + \cdots + E(Y_n)}_{n \text{ terms}}) \\
&= \frac{1}{n}(n \cdot \mu_Y) = \mu_Y
\end{aligned}$$

Note that we used the same principle that $E(Y_i) = E(Y_j) = \mu_Y$ to accomplish the above. Now to show the variance of $\bar{Y}$.

$$Var(\bar{Y}) = Var\left(\frac{1}{n}(Y_1 + Y_2 + \cdots + Y_n)\right) = \frac{1}{n^2}Var(Y_1 + Y_2 + \cdots + Y_n)$$

$$= \frac{1}{n^2}\Big(\overbrace{Var(Y_1) + Var(Y_2) + \cdots + Var(Y_n)}^{n \text{ terms}} + \overbrace{2Cov(Y_1, Y_2) + \cdots + 2Cov(Y_1, Y_n)}^{n \text{ choose 2 terms}}\Big)$$

$$= \frac{1}{n^2} \cdot n\sigma_Y^2 + \frac{1}{n^2}\binom{n}{2}2\rho\sigma_Y^2$$

$$= \frac{\sigma_Y^2}{n} + \frac{1}{n^2} \cdot \frac{n(n-1)}{2}2\rho\sigma_Y^2$$

$$= \frac{\sigma_Y^2}{n} + \frac{n-1}{n}\rho\sigma_Y^2$$

We have n choose 2 terms because this is every pairwise correlation and that's how many there exist. Otherwise, the rest is straightforward and we showcase what we are looking for.

$\square$

1d

When n is very large, show that Var$(\bar{Y}) \approx \rho\sigma_Y^2$.

Solution Attempt: Take our expression from above,

$$Var(\bar{Y}) = \frac{\sigma_Y^2}{n} + \left(\frac{n-1}{n}\right)\rho\sigma_Y^2$$

$$\frac{\sigma_Y^2}{n} \to 0 \quad \text{as} \quad n \to \infty$$

$$\frac{n-1}{n} \to 1 \quad \text{as} \quad n \to \infty$$

$Var(\bar{Y}) \to \rho\sigma_Y^2$ for large n and $Var(\bar{Y}) \approx \rho\sigma_Y^2$.

$\square$

Problem 2

On the companion website, you will find the spreadsheet `Age_HourlyEarnings`, which contains the joint distribution of age (`Age`) and average hourly earnings (`AHE`) for 25-to-34 year old full-time workers in 2012 with an education level that exceeds a high school diploma. Use this joint distribution to carry out the following exercises.

2a

Compute the marginal distribution of `Age`.

Solution Attempt: For this part and future parts, I've converted the dataset into a clean `.csv` file and saved it as `df`.

```
df.sum(axis=0)
```

```
25    0.084967
26    0.090789
27    0.085471
28    0.093518
29    0.103331
30    0.104217
31    0.103166
32    0.108084
33    0.108386
34    0.114542
dtype: float64
```

This above shows us the values of the `Age` marginal distribution. Notice that this sums to one.

```
age_marginal = df.sum(axis=0)
age_marginal.sum()
```

```
0.9964715700000001
```

□

2b

Compute the mean of `AHE` for each value of `Age`; that is, compute $E(\texttt{AHE} \mid \texttt{Age} = 25)$ and so forth.

Solution Attempt: For each age $a \in \{25, \ldots, 34\}$

$$E(\texttt{AHE} \mid \texttt{Age} = a) = \sum_y y \cdot P(\texttt{AHE} = y \mid \texttt{Age} = a) = \frac{\sum_y y \cdot P(\texttt{AHE} = y, \texttt{Age} = a)}{\sum_y P(\texttt{AHE} = y, \texttt{Age} = a)}$$

```
means = {}
for age in df.columns:
  col = df[age]
  num = (col.index * col).sum()
  den = col.sum()
  means[age] = num / den

age_means = pd.DataFrame(
  list(means.items()),
  columns = ['Age', 'E(AHE | Age)']
)
print(age_means)
```

```
   Age  E(AHE | Age)
0   25     17.634440
1   26     19.019161
2   27     19.704933
3   28     20.182865
4   29     21.142915
5   30     21.811767
6   31     22.724216
7   32     23.691860
8   33     23.279501
9   34     23.984285
```

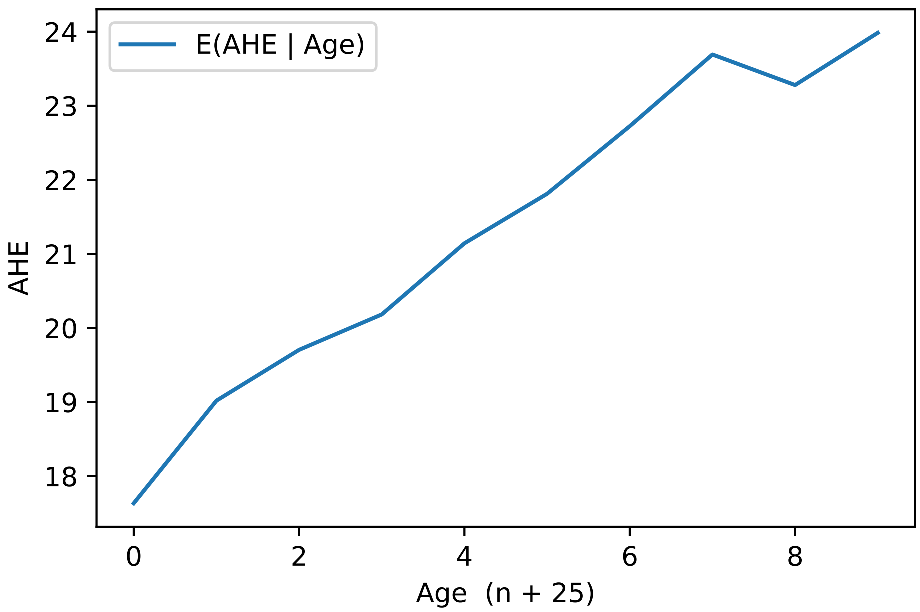
□

2c

Compute and plot the mean of `AHE` versus `Age`. Are average hourly earnings and age related? Explain.

Solution Attempt:

```
age_means.plot(
  xlabel = 'Age  (n + 25)',
  ylabel = 'AHE'
)
```



□

2d

Use the law of iterated expectations to compute the mean of AHE; that is, compute $E(\texttt{AHE})$.

Solution Attempt: Here is what we are looking for,

$$E(\texttt{AHE}) = E(E(\texttt{AHE} \mid \texttt{Age})) = \sum_a \underbrace{E(\texttt{AHE} \mid \texttt{Age} = a)}_{m(a)} \underbrace{P(\texttt{Age} = a)}_{P_A}$$

where

$$m(a) = \frac{\sigma_y y \cdot P(\texttt{AHE} = y, \texttt{Age} = a)}{\sum_y P(\texttt{AHE} = y, \texttt{Age} = a)} \quad \text{and} \quad P_a = \sum_y P(\texttt{AHE} = y, \texttt{Age} = a)$$

And, we can verify directly that

$$E(\texttt{AHE}) = \sum_a \sum_y y \cdot P(\texttt{AHE} = y, \texttt{Age} = a)$$

```
# P(Age=a)
P_age = df.sum(axis=0)

# m(a) = E[AHE | Age=a]
means = {}
for a in df.columns:
    col = df[a]
    means[a] = (col.index * col).sum() / col.sum()
m = pd.Series(means)

# E[AHE] via law of iterated expectations
E_AHE = (m * P_age).sum()
print("E[AHE] via LIE =", E_AHE)
```

```
E[AHE] via LIE = 21.430079466
```

```
E_AHE_direct = (df.mul(df.index, axis=0)).sum().sum()
print("E[AHE] direct  =", E_AHE_direct)
```

```
E[AHE] direct  = 21.430079466
```

Great, we get the same things.

□

2e

Compute the variance of `AHE`.

Solution Attempt: The direct definition,

$$\mu = E(\texttt{AHE}), \quad E(\texttt{AHE}^2) = \sum_y y^2 \cdot P(\texttt{AHE} = y), \quad \boxed{Var(\texttt{AHE}) = E(\texttt{AHE}^2) - \mu^2}$$

Law of total variance says,

$$\boxed{Var(\texttt{AHE}) = E(Var(\texttt{AHE} \mid \texttt{Age})) = Var(E(\texttt{AHE} \mid \texttt{Age}))}$$

We can do this directly, or with the law of total variance. Let's try both see if we get the same.

```python
# P(AHE = y)
P_ahe = df.sum(axis=1)

# mean and second moment
mu = (P_ahe.index * P_ahe).sum()
E2 = ((P_ahe.index**2) * P_ahe).sum()

var_ahe = E2 - mu**2
print('Var(AHE) =', var_ahe)
```

```
Var(AHE) = 165.45054164692527
```

And now with law of total variance,

```python
# weights over Age
P_age = df.sum(axis=0)

m, v = {}, {}
for a in df.columns:
  col = df[a]
  p = col / col.sum()              # P(AHE | Age=a)
  ma = (p.index * p).sum()         # E(AHE | Age=a)
  E2a = ((p.index**2) * p).sum()   # E(AHE^2 | Age=a)
  m[a] = ma
  v[a] = E2a - ma**2               # Var(AHE | Age=a)

m = pd.Series(m)
v = pd.Series(v)

mu = (m * P_age).sum()             # E(AHE)
var_ahe_LTV = (v * P_age).sum() + ((m - mu)**2 * P_age).sum()
print('Var(AHE) via LTV =', var_ahe_LTV)
```

```
Var(AHE) via LTV = 163.83011614687115
```

There seems to be some error in rounding, truncating, or data types but we get essentially the same thing. For grading purposes, go with the first one.

□

2f

Compute the covariance between `AHE` and `Age`.

Solution Attempt: Let $Y =$ `AHE` and $A =$ `Age`.

$$Cov(Y, A) = E(YA) - E(Y)E(A)$$

From the joint table $P(Y = y, A = a)$,

$$E(Y) = \sum_y y P_Y(y), \quad E(A) = \sum_a a P_A(a), \quad E(YA) = \sum_y \sum_a ya P(y, a)$$

```
# I'm going to try to fix the previous data type issue.
# Ensure numeric Age labels
df.columns = df.columns.astype(int)

# marginals
P_ahe = df.sum(1)     # P(Y)
P_age = df.sum(0)     # P(A)

# means
E_Y = (P_ahe.index * P_ahe).sum()
E_A = (P_age.index * P_age).sum()

# cross-moment E(YA) from joint
E_YA = df.mul(df.columns, 1).mul(df.index, 0).to_numpy().sum()

# covariance
cov_YA = E_YA - E_Y * E_A
print('Cov(AHE, Age) =', cov_YA)
```

```
Cov(AHE, Age) = 7.855846033653734
```

□

2g

Compute the correlation between `AHE` and `Age`.

Solution Attempt:

```
Var_Y = ((P_ahe.index ** 2) * P_ahe).sum() - E_Y ** 2
Var_A = ((P_age.index ** 2) * P_age).sum() - E_A ** 2
corr = cov_YA / (Var_Y ** 0.5 * Var_A ** 0.5)
print('Corr(AHE, Age) =', corr)
```

```
Corr(AHE, Age) = 0.18226305899108558
```

□

2h

Relate your answers in parts (f) and (g) to the plot you constructed in (c).

Solution Attempt: The plot in (c) shows $E(\texttt{AHE} \mid \texttt{Age})$ rising roughly from $18 at 25 to roughly $24 at 34, so `AHE` and `Age` (and a positive correlation, albeit not huge).

Your variance result and the law of total variance say

$$Var(\texttt{AHE}) = E(Var(\texttt{AHE} \mid \texttt{Age})) + Var(E(\texttt{AHE} \mid \texttt{Age})).$$

The plot reflects the second term (the between-age spread of the conditional means), which is small relative to the total variance— so most dispersion in `AHE` within age groups, not across ages.

So, basically, the plot explains the direction (positive association), while the variance/cov explaisn the magnitude.

□.

Problem 3

Let $X_1, \dots, X_n$ be a random sample from the Bernoulli distribution with parameter $p = 0.78$ and $n = 25$. Generate $M = 100$ such random draws (there will be $M \times n = 2500$ random variables generated). For each random draw calculate a sample mean $\bar{X}^{(j)}$, where $j = 1, \dots, M$.

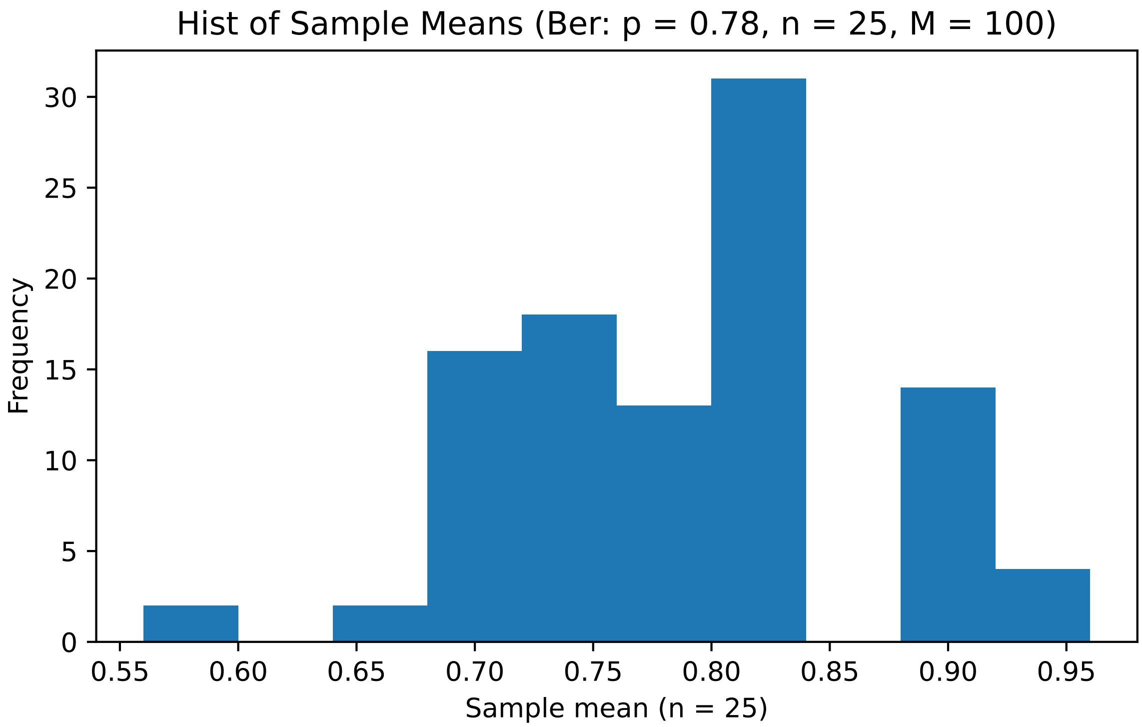
3a

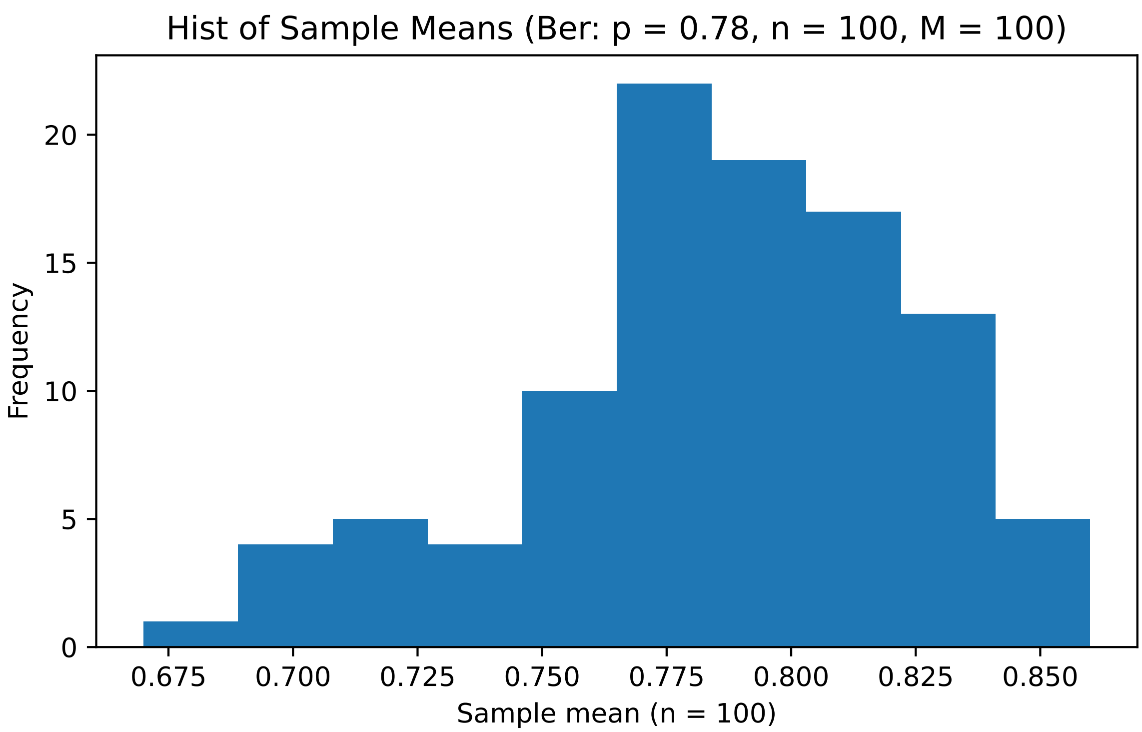
Draw a histogram $\bar{X}^{(1)}, \dots \bar{X}^{(M)}$. Repeat the same for $n = 100$. Discuss the Law of Large Numbers.

Solution Attempt:

```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# parameters
p = 0.78
M = 100
rng = np.random.default_rng(42)

def simulate_means(n, M, p, rng):
  draws = rng.binomial(1, p, size = (M, n))
  return draws.mean(1)

mean_25 = simulate_means(25, M, p, rng)
mean_100 = simulate_means(100, M, p, rng)

plt.figure(figsize = (7,4))
plt.hist(mean_25, bins = 10, )
plt.xlabel('Sample mean (n = 25)')
plt.ylabel('Frequency')
plt.title('Hist of Sample Means (Ber: p = 0.78, n = 25, M = 100)')

plt.figure(figsize = (7,4))
plt.hist(mean_100, bins = 10)
plt.xlabel('Sample mean (n = 100)')
plt.ylabel('Frequency')
plt.title('Hist of Sample Means (Ber: p = 0.78, n = 100, M = 100)')
```

```
Text(0.5, 1.0, 'Hist of Sample Means (Ber: p = 0.78, n = 100, M = 100)')
```


Hist of Sample Means (Ber: p = 0.78, n = 25, M = 100)
Frequency
Sample mean (n = 25)
0
5
10
15
20
25
30
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95


Hist of Sample Means (Ber: p = 0.78, n = 100, M = 100)
Frequency
Sample mean (n = 100)
0
5
10
15
20
0.675
0.700
0.725
0.750
0.775
0.800
0.825
0.850

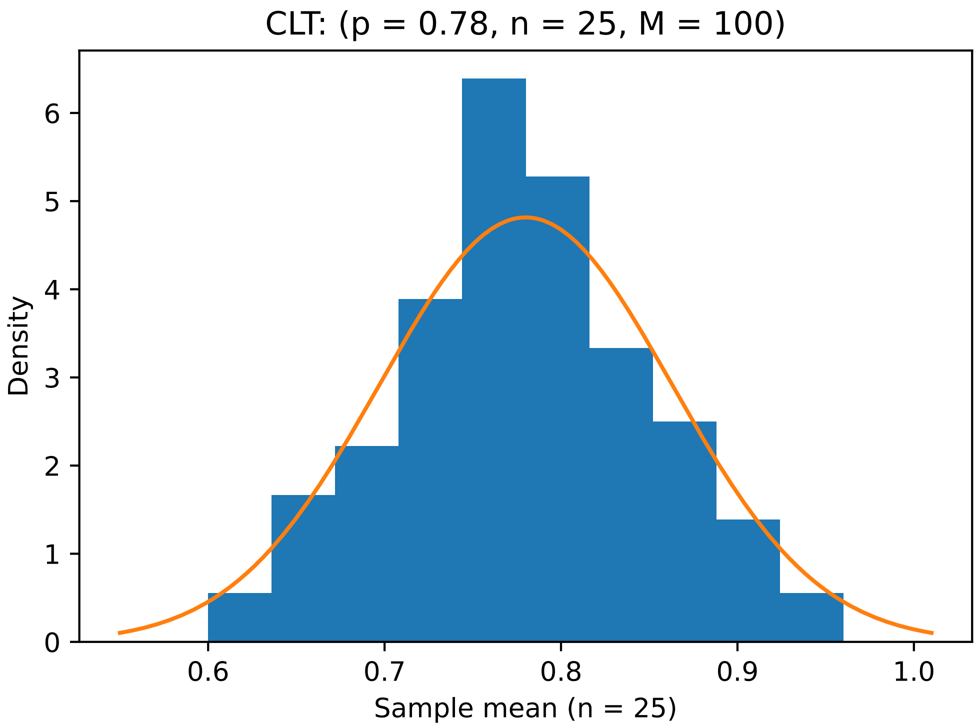
□

3b

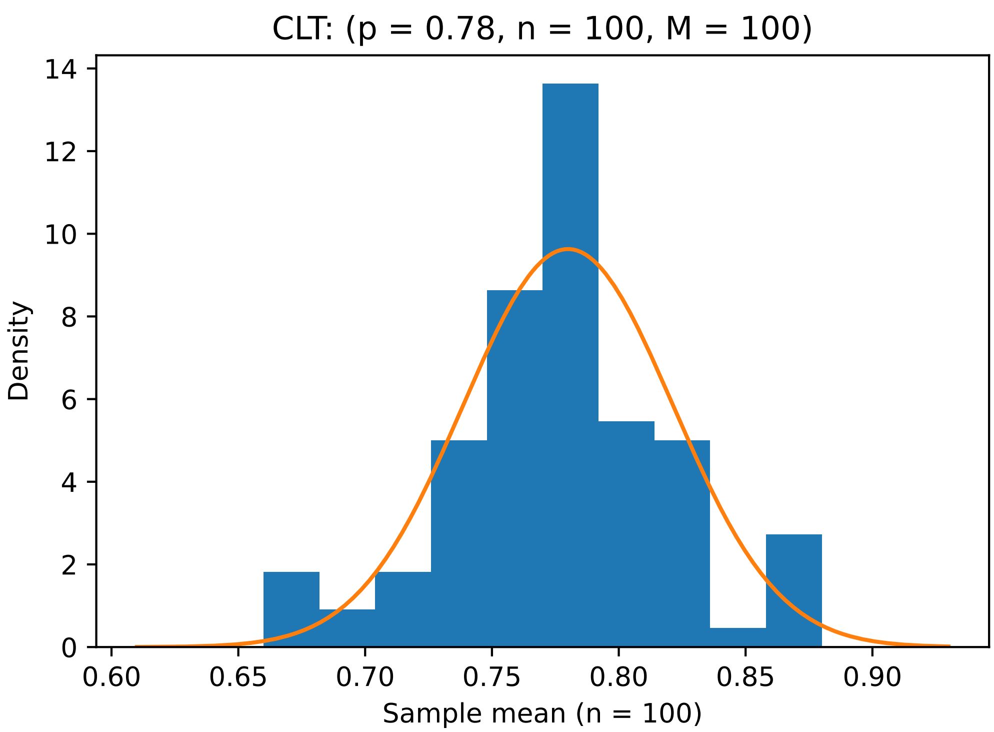
For $n = 25$ and $n = 100$ draw corresponding histograms to illustrate the Central Limit Theorem. Dicuss.

Solution Attempt:

```
from math import sqrt, pi, exp

p = 0.78
M = 100
rng = np.random.default_rng(7)

def simulate_means(n):
    x = rng.binomial(1, p, size=(M, n)).mean(1)
    return x

def normal_pdf(x, mu, sd):
    return (1 / (sd * sqrt(2 * pi))) * np.exp(-(x - mu)**2 / (2 * sd**2))

for n in (25, 100):
    means = simulate_means(n)
    mu = p
    sd = sqrt(p * (1 - p) / n)

    plt.figure(figsize = (6, 4))
    # histogram of sample means (density so total area ~1)
    plt.hist(means, bins = 10, density = True)
    # theoretical normal curve
    xs = np.linspace(min(means) - 0.05, max(means) + 0.05, 200)
    plt.plot(xs, normal_pdf(xs, mu, sd))
    plt.xlabel(f"Sample mean (n = {n})")
    plt.ylabel("Density")
    plt.title(f"CLT: (p = {p}, n = {n}, M = {M})")
```



CLT: (p = 0.78, n = 25, M = 100)
Density
Sample mean (n = 25)



CLT: (p = 0.78, n = 100, M = 100)
Density
Sample mean (n = 100)

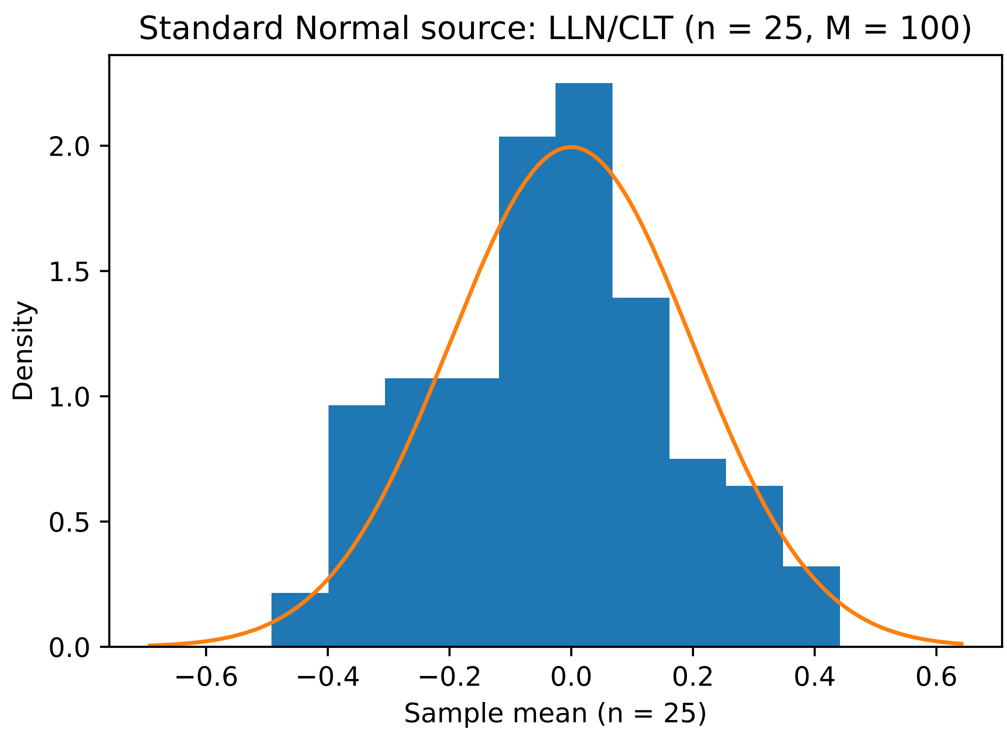
□

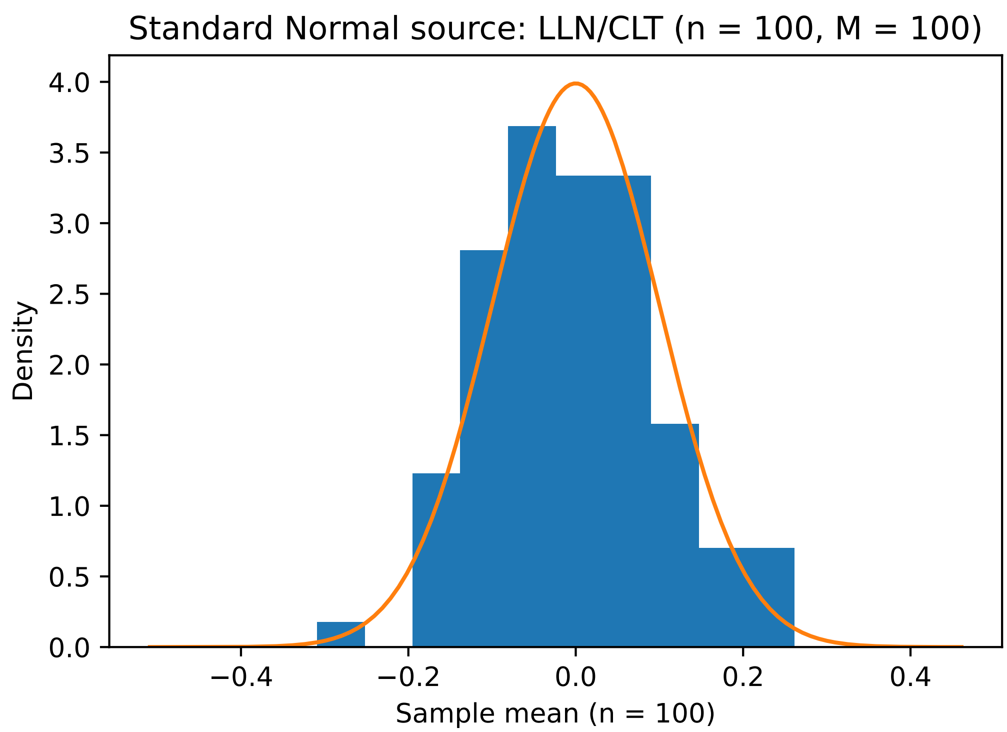
3c

Repeat (a) and (b) for the random draw from the standard normal distribution.

Solution Attempt:

```
M = 100
rng = np.random.default_rng(7)

def simulate_means_normal(n):
    # M x n draws from N(0,1); take sample means
    return rng.standard_normal(size=(M, n)).mean(axis=1)

def normal_pdf(x, mu, sd):
    return (1 / (sd * sqrt(2 * pi))) * np.exp(-(x - mu)**2 / (2 * sd**2))

for n in (25, 100):
    means = simulate_means_normal(n)
    mu, sd = 0.0, 1.0 / sqrt(n)            # exact: mean 0, Var = 1/n

    plt.figure(figsize = (6, 4))
    plt.hist(means, bins = 10, density = True)
    xs = np.linspace(means.min() - 0.2, means.max() + 0.2, 200)
    plt.plot(xs, normal_pdf(xs, mu, sd))
    plt.xlabel(f"Sample mean (n = {n})")
    plt.ylabel("Density")
    plt.title(f"Standard Normal source: LLN/CLT (n = {n}, M = {M})")
    plt.show()
```


Standard Normal source: LLN/CLT (n = 25, M = 100)
Density
2.0
1.5
1.0
0.5
0.0
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
Sample mean (n = 25)


Standard Normal source: LLN/CLT (n = 100, M = 100)
Density
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.4
−0.2
0.0
0.2
0.4
Sample mean (n = 100)

$\square$

3d

Compare your answers for the Bernoulli and the standard normal distribution for the LLN and the CLT.

Solution Attempt:

LLN (convergence of $\bar{X}$ to the true mean)

- $Ber(p = 0.78)$: $\bar{X} \to 0.78$. Spread is approx. $\sqrt{p(1-p)/n} = \sqrt{0.1716/n}$.
- Stadard Normal: $\bar{X} \to 0$. Spread is approx. $\sqrt{1/n}$.

Both concentrate at their means and shrink like $1/\sqrt{n}$. Normal shrinks from a larger variance (1 vs. 0.1716), so its histogram are wider for the same n.

CLT (shape of the sampling distribution):

- $Ber(p = 0.78)$: Approx. normal for moderate n; with $p = 0.78$ the $n = 25$ histogram can show mild skew effects (bounded in $[0, 1]$), but by $n = 100$ it's close to normal.
- Standard Normal: $\bar{X}$ is exactly $\mathcal{N}(0, 1/n)$ for any n so the curve matches quickly.

$\square$

Problem 4

4a

$\bar{Y}$ is an unbiased estimator of μ_Y. Is $\bar{Y}^2$ an unbiased estimator of μ_Y^2?

Solution Attempt: Let $Y_1, \ldots, Y_n$ be i.i.d. with $E(Y_i) = \mu_Y$ and $Var(Y_i) = \sigma_Y^2$. Then

$$E(\bar{Y}) = \mu_Y \quad \text{and} \quad Var(\bar{Y}) = \frac{\sigma_Y^2}{n}.$$

Using $E(X^2) = Var(X) + (E(X))^2$,

$$E(\bar{Y}^2) = Var(\bar{Y}) + \big(E(\bar{Y})\big)^2 = \frac{\sigma_Y^2}{n} + \mu_Y^2.$$

Hence

$$E(\bar{Y}^2) - \mu_Y^2 = \frac{\sigma_Y^2}{n} \neq 0$$

unless $\sigma_Y^2 = 0$ (degenerate case). Therefore, $\bar{Y}^2$ is *not* an unbiased estimator of μ_Y^2 (it is biased upward by σ_Y^2/n). It is, however, consistent since the bias $\to 0$ as $n \to \infty$.

$\square$

4a

$\bar{Y}$ is an unbiased estimator of μ_Y. Is $\bar{Y}^2$ an unbiased estimator of μ_Y^2?

Solution Attempt: No. Both denote the same quantity, the square of the mean: $\mu_Y^2 = (\mu_Y)^2$. The notation change is cosmetic and does not alter the mathematics.

$\square$

Problem 5

Show that the sample covariance between X and Y can be defined by

$$\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$$

or

$$\frac{1}{n-1}\sum_{i=1}^{n} X_i Y_i - \frac{n}{n-1}\,\bar{X}\,\bar{Y}.$$

Solution Attempt: Expand and simplify:

$$\begin{aligned}
\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}) &= \sum_{i=1}^{n}\left(X_iY_i - \bar{X}Y_i - \bar{Y}X_i + \bar{X}\bar{Y}\right) \\
&= \sum_{i=1}^{n} X_iY_i - \bar{X}\sum_{i=1}^{n} Y_i - \bar{Y}\sum_{i=1}^{n} X_i + n\bar{X}\bar{Y} \\
&= \sum_{i=1}^{n} X_iY_i - \bar{X}\,(n\bar{Y}) - \bar{Y}\,(n\bar{X}) + n\bar{X}\bar{Y} \\
&= \sum_{i=1}^{n} X_iY_i - n\bar{X}\bar{Y}.
\end{aligned}$$

Divide both sides by $n-1$ to obtain

$$\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}) = \frac{1}{n-1}\sum_{i=1}^{n} X_iY_i - \frac{n}{n-1}\,\bar{X}\,\bar{Y},$$

which shows the two formulas are equivalent.

$\square$

Problem 6

Grades on a standardized test are known to have a mean of 1000 for students in the United States. The test is administered to 453 randomly selected students in Florida; in this sample, the mean is 1013, and the standard deviation is 108.

6a

Construct a 95% confidence interval for the average test score for Florida students.

Solution Attempt: Use a one-sample t CI (large n so $t \approx 1.96$): $\bar{X} \pm t_{0.975}\, s/\sqrt{n}$. Here, $\bar{X} = 1013$, $s = 108$, $n = 453$, so $SE = s/\sqrt{n} = 108/\sqrt{453} \approx 5.074$. Thus the 95% CI is

$$1013 \pm 1.96(5.074) \;=\; 1013 \pm 9.946 \;=\; (1003.05,\; 1022.95).$$

□

6b

Is there statistically significant evidence that Florida students perform differently than other students in the United States?

Solution Attempt: Test $H_0 : \mu = 1000$ vs. $H_1 : \mu \neq 1000$. Test statistic

$$z \approx \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = \frac{1013 - 1000}{108/\sqrt{453}} \approx \frac{13}{5.074} \approx 2.562.$$

Two-sided p–value $\approx 0.0104 < 0.05$, so there is statistically significant evidence that Florida's mean differs from 1000.

□

6c

Another 503 students are selected at random from Florida. They are given a 3-hour preparation course before the test is administered. Their average test score is 1019, with a standard deviation of 95.

6c (i)

Construct a 95% confidence interval for the change in the average test score associated with the prep course.

Solution Attempt: Treat as two independent samples. Let $\Delta = \bar{X}_{\text{prep}} - \bar{X}_{\text{orig}} = 1019 - 1013 = 6$. The standard error is

$$SE_\Delta = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{108^2}{453} + \frac{95^2}{503}} \approx 6.610.$$

A 95% CI: $\Delta \pm 1.96\, SE_\Delta = 6 \pm 1.96(6.610) = 6 \pm 12.97 = (-6.97,\ 18.97)$.

□

6c (ii)

Is there statistically significant evidence that the prep course helped?

Solution Attempt: Test $H_0 : \Delta = 0$ vs. $H_1 : \Delta > 0$. Test statistic $t = \Delta / SE_\Delta = 6/6.610 \approx 0.908$. One-sided $p \approx 0.18$ (two-sided ≈ 0.36). Not statistically significant at 5%.

□

6d

The original 453 students are given the prep course and then are asked to take the test a second time. The average change in their test scores is 9 points, and the standard deviation of the change is 60 points.

6d (i)

Construct a 95% confidence interval for the change in average test scores.

Solution Attempt: Paired setting on the same students. Let $\bar{D} = 9$, $s_D = 60$, $n = 453$, $SE_D = s_D/\sqrt{n} = 60/\sqrt{453} \approx 2.819$. 95% CI: $\bar{D} \pm 1.96\, SE_D = 9 \pm 1.96(2.819) = 9 \pm 5.525 = (3.48,\ 14.53)$.

□

6d (ii)

Is there statistically significant evidence that the students will perform better on their second attempt, after taking the prep course?

Solution Attempt: One-sided test $H_0 : \mu_D \leq 0$ vs. $H_1 : \mu_D > 0$. $t = \bar{D}/SE_D = 9/2.819 \approx 3.193$; one-sided $p \approx 0.0007 < 0.05$. Yes, there is significant evidence of improvement.

□

6d (iii)

Students may have performed better in their second attempt because of the prep course or because they gained test-taking experience in their first attempt. Describe an experiment that would quantify these two effects.

Solution Attempt: We can run a 2 by 2 experiment with prep (yes/no) and practice (first/retake). Randomly assign students to four groups.

- Prep before first test, no retake (measures pure prep effect)
- No prep before first test, no retake (base)
- No prep before first test, retake without prep (measures practice)
- Prep before retake (adds prep effect on top of practice)

Then we analyze with a two-way ANOVA

- Practice effect = score gain from retaking without prep.
- Prep effect = score gap between prepped vs. unprepped at the same test number. Making all this random makes sure that we measure prep and practice separately.

□

Problem 7

Math SAT score (Y) are normally distributed with a mean of 500 and a standard deviation of 100. An evening school advertises that it can improve students' scores by roughly a third of a standard deviation, or 30 points, if they attend a course which runs over several weeks. The statistician for a consumer protection agency suspects that the courses are not effective. She views the situation as follows: $H_0 : \mu_Y = 500$ versus $H_1 : \mu_Y = 530$.

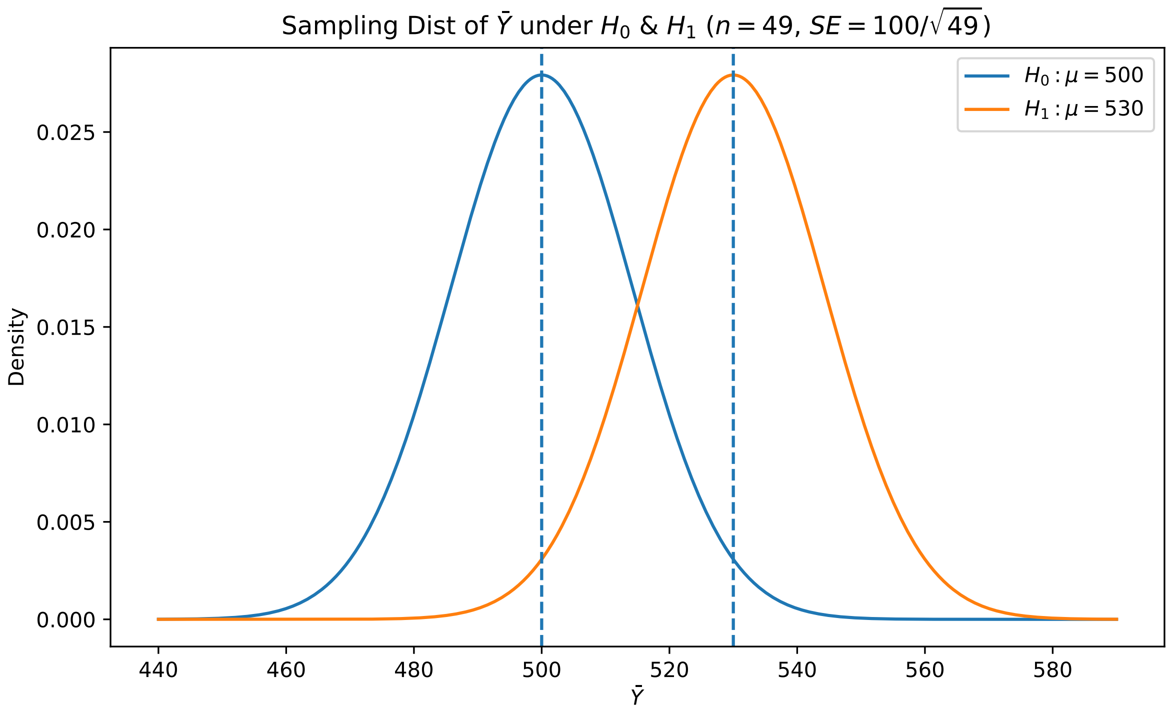
7a

Sketch the two distributions under the null hypothesis and the alternative hypothesis.

Solution Attempt: The relevant curves are the *sampling distributions* of $\bar{Y}$ (for the test). With $n = 49$, $SE = 100/\sqrt{49} = 100/7 \approx 14.286$. So

$$\bar{Y} \mid H_0 \sim \mathcal{N}\Big(500,\ (100/\sqrt{49})^2\Big), \qquad \bar{Y} \mid H_1 \sim \mathcal{N}\Big(530,\ (100/\sqrt{49})^2\Big).$$

```python
import numpy as np
import matplotlib.pyplot as plt

mu0 = 500
mu1 = 530
sigma = 100
n = 49
se = sigma / np.sqrt(n)

x = np.linspace(440, 590, 400)
f0 = (1/(se*np.sqrt(2*np.pi))) * np.exp(-(x-mu0)**2/(2*se**2))
f1 = (1/(se*np.sqrt(2*np.pi))) * np.exp(-(x-mu1)**2/(2*se**2))

plt.figure(figsize=(8,5))
plt.plot(x, f0, label=r'$H_0: \mu=500$')
plt.plot(x, f1, label=r'$H_1: \mu=530$')
plt.axvline(mu0, linestyle='--')
plt.axvline(mu1, linestyle='--')
plt.title('Sampling Dist of $\\bar{Y}$ under $H_0$ & $H_1$ ($n=49$, $SE=100/\\sqrt{49}$)')
plt.xlabel(r'$\bar{Y}$')
plt.ylabel('Density')
plt.legend()
plt.tight_layout()
plt.show()
```



(Plot shows two normals centered at 500 and 530 with the same spread $SE = 100/\sqrt{49}$.)

□

7b

The consumer protection agency wants to evaluate this claim by sending 50 students to attend classes. One of the students becomes sick during the course and drops out. What is the distribution of the average score of the remaining 49 students under the null, and under the alternative hypothesis?

Solution Attempt: By normal theory,

$$\bar{Y} \mid H_0 \sim \mathcal{N}\Big(500,\ (100/\sqrt{49})^2\Big), \qquad \bar{Y} \mid H_1 \sim \mathcal{N}\Big(530,\ (100/\sqrt{49})^2\Big).$$

□

7c

Assume that after graduating from the course, the 49 participants take the SAT test and score an average of 520. Is this convincing evidence that the school has fallen short of its claim? What is the p-value for such a score under the null hypothesis?

Solution Attempt: Use a one–sided test that rejects for large $\bar{Y}$ (improvement).

$$z = \frac{520 - 500}{100/\sqrt{49}} = \frac{20}{14.286} \approx 1.40.$$

p–value (upper tail) $= 1 - \Phi(1.40) \approx 0.081$. Not significant at 5%; the data do **not** provide convincing evidence of a 30-point improvement.

$\square$

7d

What would be the critical value under the null hypothesis if the size of your test were 5%?

Solution Attempt: One–sided $\alpha = 0.05$: critical value

$$c = 500 + z_{0.95} \cdot \frac{100}{\sqrt{49}} = 500 + 1.6449 \cdot 14.286 \approx 523.50.$$

Reject H_0 if $\bar{Y} > 523.50$.

$\square$

7e

Given this critical value, what is the power of the test? What options does the statistican have for increasing the power in this situation?

Solution Attempt: Power under H_1 is

$$P_{H_1}(\bar{Y} > c) = 1 - \Phi\left(\frac{c - 530}{100/\sqrt{49}}\right) = 1 - \Phi\left(\frac{523.50 - 530}{14.286}\right) = 1 - \Phi(-0.455) \approx \Phi(0.455) \approx 0.675.$$

So the power is about 67.5%. To increase power: increase n (largest effect), use a larger α, reduce variance (more reliable measurement), or ensure a one-sided test aligned with the claim.

$\square$

Problem 8

Assume that under the null hypothesis, $\bar{Y}$ has an expected value of 500 and a standard deviation of 20. Under the alternative hypothesis, the expected value is 550. Sketch the probability density function for the null and alternative hypothesis in the same figure. Pick a critical value such that the p-value is approximately 5%. Mark the areas, which show the size and the power of the test. What happens to the power of the test if the altnernative hypothesis moves close to the null hypothesis, i.e. $\mu_y = 540, 530, 520$?

Solution Attempt: I am going to try and plot this with `matplotlib`.

Imports,

```
import matplotlib.pyplot as plt
from math import erf, sqrt
```

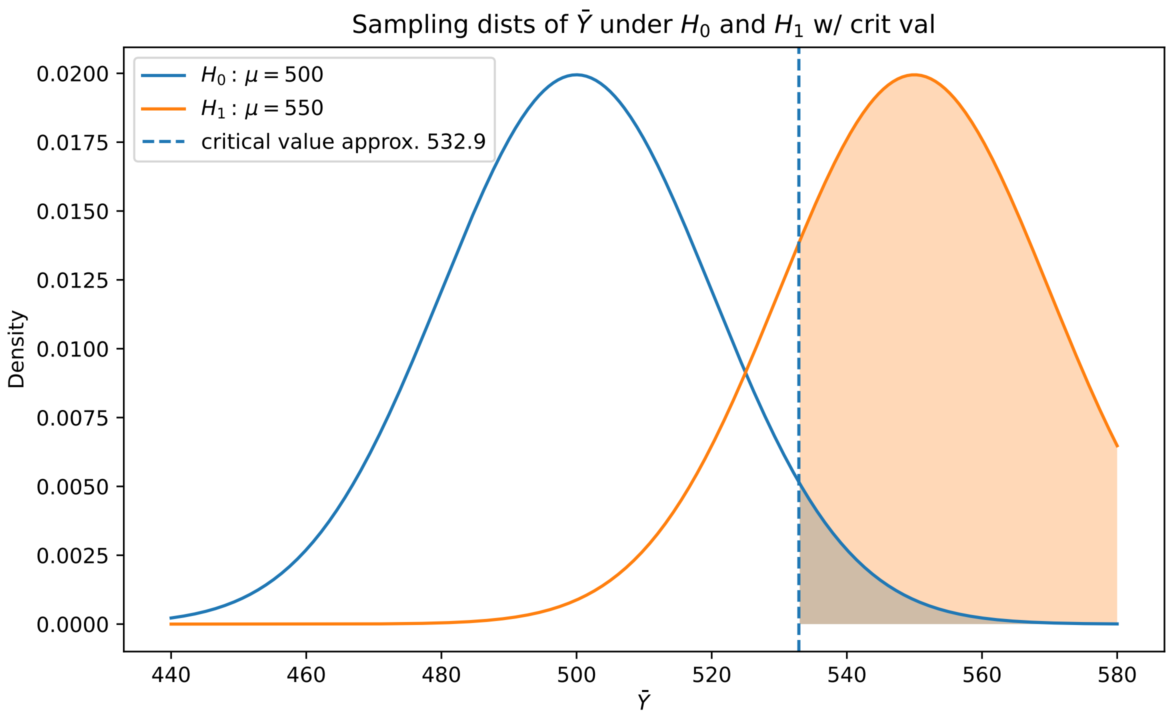
Parameters and functions,

```
# Given
mu0 = 500
sigma = 20  # SD under both hypotheses (sampling distribution of \bar Y)
mu1 = 550   # primary alternative
alpha = 0.05

# Critical value for one-sided upper test at ~5%
z_alpha = 1.6448536269514722  # Phi^{-1}(0.95)
c = mu0 + z_alpha * sigma  # critical value

# x-range for plotting
x = np.linspace(440, 580, 600)
def normal_pdf(x, mu, sd):
    return (1/(sd*np.sqrt(2*np.pi))) * np.exp(-(x-mu)**2/(2*sd**2))

f0 = normal_pdf(x, mu0, sigma)
f1 = normal_pdf(x, mu1, sigma)
```

The first plot,

```
# null vs alternative (550) with shaded size and power
plt.figure(figsize=(8,5))
plt.plot(x, f0, label=r'$H_0:\, \mu=500$')
plt.plot(x, f1, label=r'$H_1:\, \mu=550$')
```

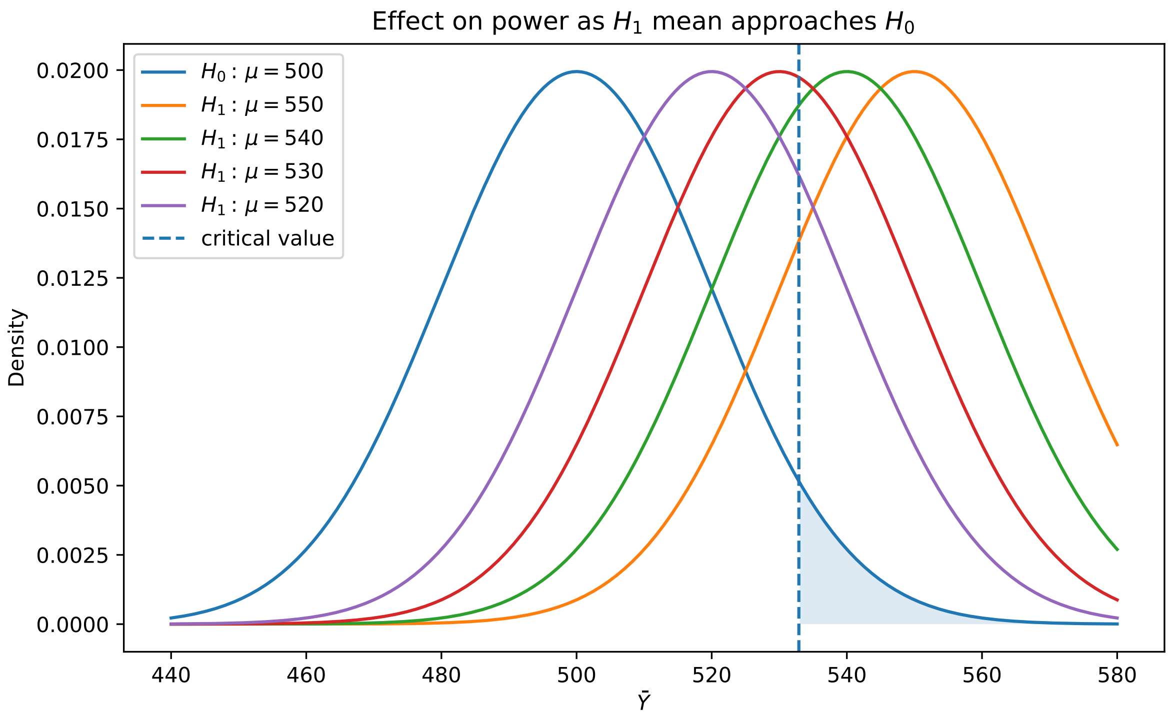
```python
plt.axvline(c, linestyle='--', label=f'critical value approx. {c:.1f}')

# Shade alpha under H0 to the right of c
mask_alpha = x >= c
plt.fill_between(x[mask_alpha], f0[mask_alpha], alpha=0.3)

# Shade power under H1 to the right of c
plt.fill_between(x[mask_alpha], f1[mask_alpha], alpha=0.3)

plt.title('Sampling dists of $\\bar{Y}$ under $H_0$ and $H_1$ w/ crit val')
plt.xlabel(r'$\bar{Y}$')
plt.ylabel('Density')
plt.legend()
plt.tight_layout()
plt.show()
```



The second plot,

```python
# power comparison as alternative mean moves closer
mus = [550, 540, 530, 520]
plt.figure(figsize=(8,5))
```

```python
plt.plot(x, f0, label=r'$H_0:\, \mu=500$')
for m in mus:
    plt.plot(x, normal_pdf(x, m, sigma), label=rf'$H_1:\, \mu={m}$')
plt.axvline(c, linestyle='--', label='critical value')
mask = x >= c
plt.fill_between(x[mask], normal_pdf(x[mask], mu0, sigma), alpha=0.15)
plt.title('Effect on power as $H_1$ mean approaches $H_0$')
plt.xlabel(r'$\bar{Y}$')
plt.ylabel('Density')
plt.legend()
plt.tight_layout()
plt.show()

# Compute powers numerically for the listed alternatives

def Phi(z):
    return 0.5*(1 + erf(z/np.sqrt(2)))

powers = {}
for m in mus:
    # Power = P_{mu=m}(\bar{Y} > c) = 1 - Phi((c - m)/sigma)
    z = (c - m)/sigma
    power = 1 - Phi(z)
    powers[m] = power

powers
```




```
{550: 0.8037649400154944,
 540: 0.6387600313123358,
 530: 0.4424132202501243,
 520: 0.2595110228414448}
```

Use a one-sided upper test (reject for large $\bar{Y}$). With $\sigma_{\bar{Y}} = 20$ and $\alpha \approx 0.05$, the critical value is

$$c \;=\; 500 + z_{0.95} \cdot 20 \;=\; 500 + 1.6449 \cdot 20 \;\approx\; 532.9.$$

Size (Type I error) is the right–tail area under H_0 beyond c; this is ~5%.

Power at μ is $1 - \Phi\big((c-\mu)/20\big)$, the right–tail area under the alternative beyond c.

Numerical power:

$$\begin{aligned}
\mu = 550: &\quad 1 - \Phi\big((532.9 - 550)/20\big) \approx 0.804,\\
\mu = 540: &\quad 1 - \Phi\big((532.9 - 540)/20\big) \approx 0.639,\\
\mu = 530: &\quad 1 - \Phi\big((532.9 - 530)/20\big) \approx 0.442,\\
\mu = 520: &\quad 1 - \Phi\big((532.9 - 520)/20\big) \approx 0.260.
\end{aligned}$$

As the alternative mean moves closer to the null (550 $\to$ 540 $\to$ 530 $\to$ 520), the two sampling distributions overlap more, so the shaded power region shrinks—power decreases.

$\square$

Problem 9

Some policy advisors have argued that education should be subsidized in developing countries to reduce fertility rates. To investigate whether or not education fertility are correlated, you collect data on population growth rates (Y) and education (X) for 86 countries. Given the sums below, compute the sample correlation:

$$\sum_{i=1}^{n} Y_i = 1.594;\ \sum_{i=1}^{n} X_i = 449.6;\ \sum_{i=1}^{n} Y_i X_i = 6.4697;\ \sum_{i=1}^{n} Y_i^2 = 0.03982;\ \sum_{i=1}^{n} X_i^2 = 3022.76$$

Solution Attempt: Use

$$r = \frac{S_{XY}}{\sqrt{S_{XX} S_{YY}}}, \quad S_{XY} = \sum X_i Y_i - \frac{\sum X_i \sum Y_i}{n},\ S_{XX} = \sum X_i^2 - \frac{(\sum X_i)^2}{n},\ S_{YY} = \sum Y_i^2 - \frac{(\sum Y_i)^2}{n}.$$

$$\begin{aligned}
S_{XY} &= 6.4697 - \frac{(449.6)(1.594)}{86} = 6.4697 - 8.3332837209 \approx -1.863583721, \\
S_{XX} &= 3022.76 - \frac{(449.6)^2}{86} \approx 3022.76 - 2350.4669767 \approx 672.2930233, \\
S_{YY} &= 0.03982 - \frac{(1.594)^2}{86} \approx 0.03982 - 0.0295446047 \approx 0.0102753953.
\end{aligned}$$

Therefore

$$r = \frac{-1.863583721}{\sqrt{(672.2930233)(0.0102753953)}} \approx -0.709.$$

The sample correlation is about $r \approx -0.709$ (negative association: higher education is associated with lower population growth).

$\square$

Problem 10

Let X be a Bernoulli random variable with parameter p. Suppose you observe a random sample $(X_i)_{i=1}^n$.

10a

Denote the parameter of interest as $\theta \equiv p^3$. Propose a consistent estimator for θ. Why is it consistent?

Solution Attempt: A natural estimator is

$$\hat{\theta} = (\bar{X})^3, \qquad \bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

By the Law of Large Numbers, $\bar{X} \xrightarrow{p} p$. By the Continuous Mapping Theorem, $(\bar{X})^3 \xrightarrow{p} p^3 = \theta$, so $\hat{\theta}$ is consistent.

(Alternative: the U–statistic $\tilde{\theta} = \binom{n}{3}^{-1} \sum_{i<j<k} X_i X_j X_k$ is unbiased for p^3 and also consistent.)

□

10b

Let $\hat{\theta}$ the estimator you proposed in part (a). Describe $E(\hat{\theta})$. Is your estimator unbiased? *Hint: you may need to use formula:*

$$\Big(\sum_{i=1}^{n} X_i\Big)^3 = \sum_{i=1}^{n} X_i^3 + 3\sum_{i \neq j} X_i X_j^2 + 6 \sum_{i \neq j \neq \ell} X_i X_j X_\ell.$$

Solution Attempt: Let $S = \sum_{i=1}^{n} X_i \sim \text{Bin}(n, p)$. Then $\hat{\theta} = (\bar{X})^3 = (S/n)^3$, so

$$E(\hat{\theta}) = \frac{E(S^3)}{n^3}.$$

For Bernoulli $X_i^k = X_i$ and independence gives $E(X_i) = p$, $E(X_i X_j) = p^2$ $(i \neq j)$, $E(X_i X_j X_\ell) = p^3$ $(i \neq j \neq \ell)$. Counting terms,

$$E(S^3) = np \; + \; 3n(n-1)p^2 \; + \; n(n-1)(n-2)p^3.$$

Hence

$$E(\hat{\theta}) = \frac{np}{n^3} + \frac{3n(n-1)p^2}{n^3} + \frac{n(n-1)(n-2)p^3}{n^3} = \frac{p}{n^2} + \frac{3(n-1)}{n^2}p^2 + \frac{(n-1)(n-2)}{n^2}p^3.$$

Therefore $\hat{\theta} = (\bar{X})^3$ is **biased** for finite n (since $E[\hat{\theta}] \neq p^3$ in general), but it is **asymptotically unbiased and consistent** because $E[\hat{\theta}] \to p^3$ as $n \to \infty$.

□

Problem 11

Assume you wanted to compare the expected values of two random variables $E(X)$ and $E(Y)$.

11a

How would you test the null hypothesis $H_0 : E(X) = E(Y)$ against the alternative $H_1 : E(X) \neq E(Y)$ if you observe two independent random samples $(X_i)_{i=1}^{n_1}$ and $(Y_j)_{j=1}^{n_2}$?

Solution Attempt:

11b

How would you test the null hypothesis $H_0 : E(X) = E(Y)$ against the alternative hypothesis $H_1 : E(X) \neq E(Y)$ if you observe one random sample $(X_i, Y_i)_{i=1}^{n}$?

Solution Attempt:

Problem 12

Suppose you are interested in a parameter $\theta > 0$ and you have a random sample $(X_i)_{i=1}^{n}$ (note that the parameter θ is strictly positive, like a variance for example). Suppose $\hat{Q}$ is a statistic and suppose that the distribution of X is such that

$$n \cdot \left(\frac{\hat{Q}}{\theta}\right)^2 \sim F, \quad \text{where} \quad F(z) = \begin{cases} 0, & z < 0 \\ 1 - e^{-z/2}, & z \in [0, \infty). \end{cases}$$

F is a special case of the Exponential Distribution (equivalently, χ^2_2).

Suppose you want to test the null hypothesis $H_0 : \theta = \theta_0$ against the alternative hypothesis $H_1 : \theta < \theta_0$, and that for this purpose you want to use the test-statistic

$$T = n \cdot \left(\frac{\widehat{Q}}{\theta_0}\right)^2$$

and a rejection rule "reject H_0 if and only if $T < c$", where c is a critical value. Let $\alpha \in (0, 1)$ denote the target significance level for the test.

12a

Find the critical value c you need to use to attain the target significance level α.

Solution Attempt: Under H_0, $T \sim F$ with $F(t) = 1 - e^{-t/2}$ for $t \geq 0$. We need

$$\alpha = P_{H_0}(T < c) = F(c) = 1 - e^{-c/2} \quad \Longrightarrow \quad e^{-c/2} = 1 - \alpha \quad \Longrightarrow \quad c = -2\ln(1-\alpha).$$

□

12b

Let t denote the value of T in your sample. Find an expression for the p-value for the test. Suppose $\theta_0 = 20$. Compute the p-value of the test when $n = 100$ and $\widehat{Q} = 0.5$.

Solution Attempt: For a lower–tail test, the p–value is

$$p\text{-value} \;=\; P_{H_0}(T \leq t) \;=\; F(t) \;=\; 1 - e^{-t/2}.$$

With $\theta_0 = 20$, $n = 100$, $\widehat{Q} = 0.5$,

$$t \;=\; n\Big(\frac{\widehat{Q}}{\theta_0}\Big)^2 \;=\; 100\Big(\frac{0.5}{20}\Big)^2 \;=\; 100 \cdot 0.000625 \;=\; 0.0625,$$

so

$$p\text{-value} \;=\; 1 - e^{-0.0625/2} \;=\; 1 - e^{-0.03125} \;\approx\; 0.0308.$$

□

12c

Find the power of the test when the true value of θ is some θ^*, with $\theta^* \neq \theta_0$. Suppose $\theta_0 = 20$, $n = 100$, and $\widehat{Q} = 0.5$. Compute the power of the test when $\theta^* = 10$.

Solution Attempt: If the true value is θ^*, then

$$U \;=\; n\Big(\frac{\widehat{Q}}{\theta^*}\Big)^2 \sim \chi^2_2, \qquad T \;=\; n\Big(\frac{\widehat{Q}}{\theta_0}\Big)^2 \;=\; \Big(\frac{\theta^*}{\theta_0}\Big)^2 U.$$

Thus the power at θ^* is

$$P_{\theta^*}(T < c) \;=\; P\bigg(U < \frac{c}{(\theta^*/\theta_0)^2}\bigg) \;=\; F\bigg(\frac{c}{(\theta^*/\theta_0)^2}\bigg) \;=\; 1 - \exp\bigg(-\frac{c}{2}\Big(\frac{\theta_0}{\theta^*}\Big)^2\bigg).$$

For a concrete number, take the usual $\alpha = 0.05$ so $c = -2\ln(1 - 0.05) \approx 0.1026$, and $\theta^* = 10$ with $\theta_0 = 20$ gives $(\theta_0/\theta^*)^2 = 4$. Then

$$\text{Power} \;=\; 1 - \exp\Big(-\frac{0.1026}{2}\cdot 4\Big) \;=\; 1 - e^{-0.2052} \;\approx\; 0.186.$$

(Note: power depends on α via c, and on θ^*; it does not depend on the realized $\widehat{Q}$.)

□

Problem 13

Suppose $Y_i \sim$ i.i.d. $\mathcal{N}(\mu_Y, \sigma_Y^2)$ for $i = 1, \ldots, n$ with σ_Y^2 known. Find t-statistics for testing $H_0 : \mu_Y = 0$ versus $H_1 : \mu_Y > 0$. Suppose $\sigma_Y = 10$ and $n = 100$. Using a test with size 5% and the corresponding rejection rule, answer the following.

Setup, With known variance, use

$$t \;=\; \frac{\bar{Y} - 0}{\sigma_Y/\sqrt{n}} \;=\; \frac{\bar{Y}}{10/\sqrt{100}} \;=\; \frac{\bar{Y}}{1}, \quad t \mid H_0 \sim \mathcal{N}(0,1).$$

At size 5% (one-sided), reject H_0 if $t > 1.64$ (critical value $z_{0.95} \approx 1.64$).

13a

Suppose $\mu_Y = 0$, so the null hypothesis is true. What is the probability that the null hypothesis is rejected?

Solution Attempt: Under H_0, $t \sim \mathcal{N}(0,1)$, so

$$P(\text{reject} \mid H_0) = P(t > 1.64) = 1 - \Phi(1.64) \approx 0.0505 \approx 5\%.$$

□

13b

Suppose $\mu_Y = 2$, so the alternative hypothesis is true. What is the probability that the null hypothesis is rejected?

Solution Attempt: Under $\mu_Y = 2$, $t = \bar{Y}/1 \sim \mathcal{N}(2,1)$, hence

$$P(\text{reject} \mid \mu_Y = 2) = P(t > 1.64) = 1 - \Phi(1.64 - 2) = 1 - \Phi(-0.36) = \Phi(0.36) \approx 0.641.$$

□

13c

Suppose that in 90% of cases the data are drawn from a population where the null is true ($\mu_Y = 0$) and in the 10% of cases the data came from a population where the alternative is true and $\mu_Y = 2$.

13c (i)

You complete the t-statistics. What is the probability that $t > 1.64$, that is, that you reject the null hypothesis?

Solution Attempt: By the law of total probability,

$P(t > 1.64) = 0.9\, P(t > 1.64 \mid \mu = 0) + 0.1\, P(t > 1.64 \mid \mu = 2) = 0.9(0.0505) + 0.1(0.6406) \approx 0.110.$

$\square$

13c (ii)

Suppose you reject the null hypothesis; that is $t > 1.64$. What is the probability that the sample data were drawn from the $mu_y = 0$ population?

Solution Attempt: By Bayes' rule,

$$P(\mu = 0 \mid t > 1.64) = \frac{0.9\, P(t > 1.64 \mid \mu = 0)}{P(t > 1.64)} = \frac{0.9 \cdot 0.0505}{0.110} \approx 0.415.$$

$\square$

13d

It is hard to discover a new effective drug. Suppose 90% of new drugs are ineffective and only 10% are effective. Let Y denote the drop in the level of a specific blood toxin for a patient taking a new drug. If the drug is ineffective, $\mu_Y = 0$ and $\sigma_Y = 10$; if the drug is effective, $\mu_Y = 2$ and $\sigma_Y = 10$.

13d (i)

A new drug is tested on a random sample of $n = 100$ patients, data are collected, and the resulting t-statistics is found to be greater than 1.64. What is the probability that the drug is ineffective?

Solution Attempt: Same mixture as 13c with priors $0.9/0.1$ and the same tail probabilities, so

$$P(\text{ineffective} \mid t > 1.64) \approx 0.415.$$

$\square$

13d (ii)

Suppose the one-sided test uses the 0.5% significance level. What is the probability the drug is ineffective?

Solution Attempt: Use critical value $z_{0.995} \approx 2.575$. Then

$$P(t > 2.575 \mid \mu = 0) = 0.005, \qquad P(t > 2.575 \mid \mu = 2) = 1 - \Phi(2.575 - 2) = 1 - \Phi(0.575) \approx 0.282.$$

Bayes update:

$$P(\text{ineffective} \mid t > 2.575) = \frac{0.9 \cdot 0.005}{0.9 \cdot 0.005 + 0.1 \cdot 0.282} = \frac{0.0045}{0.0045 + 0.0282} \approx 0.138.$$

$\square$